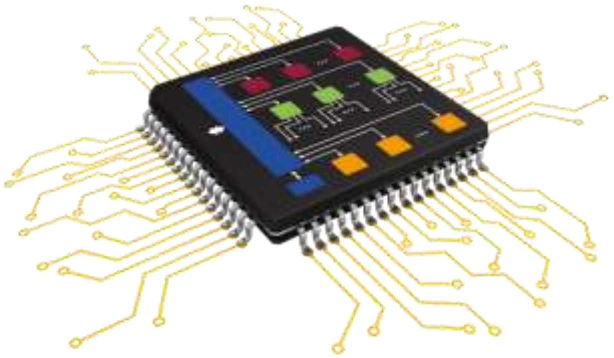
Applications of Embedded Systems



WHAT IS AN EMBEDDED SYSTEM?

Embedded : means hidden inside so one can't see it

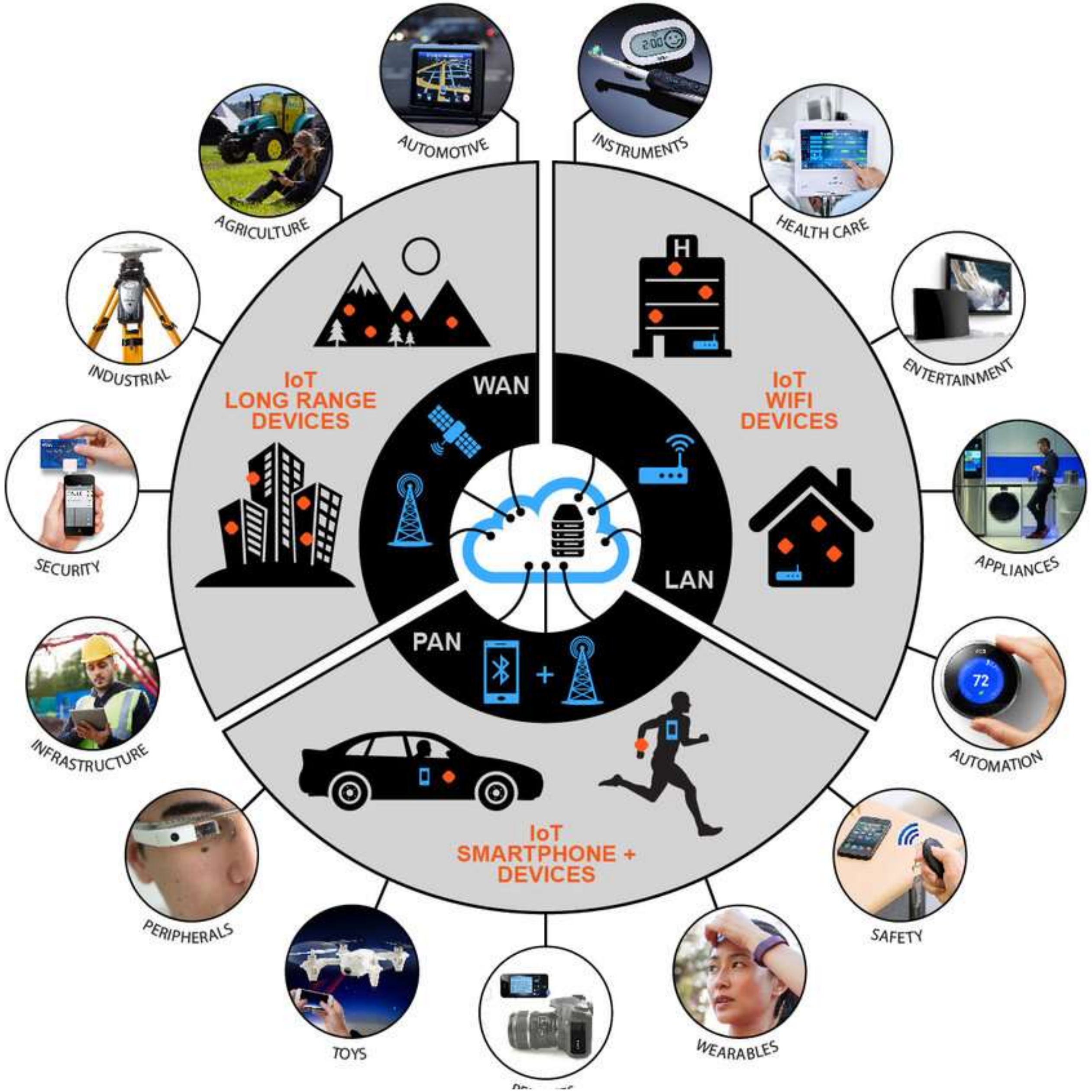
System : means multiple components interfaced together for doing special purpose .

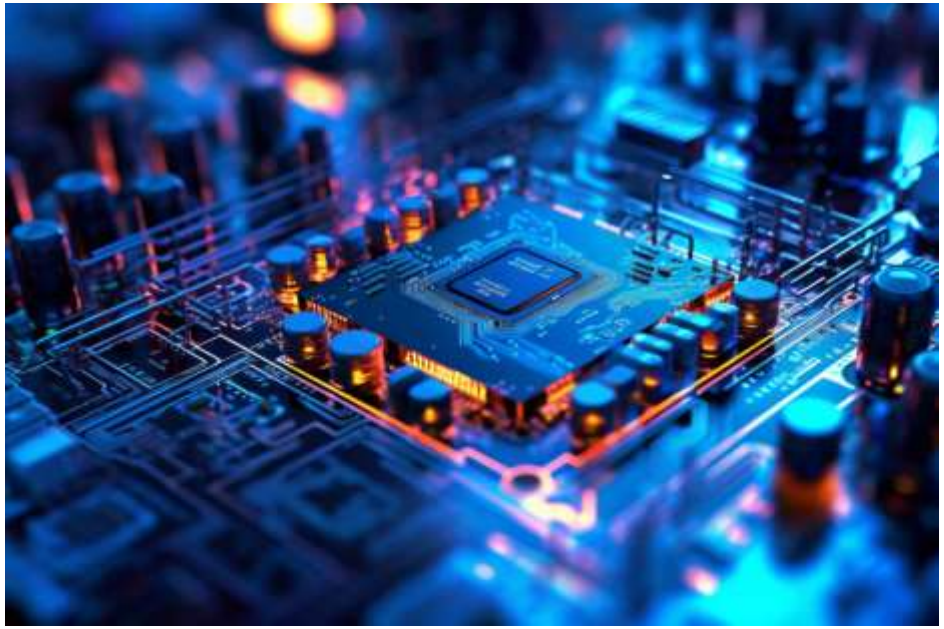
Embedded System : is a special-purpose computer system designed to perform certain dedicated functions.

Functionalities : are done by dedicated HW and SW with limited resources.

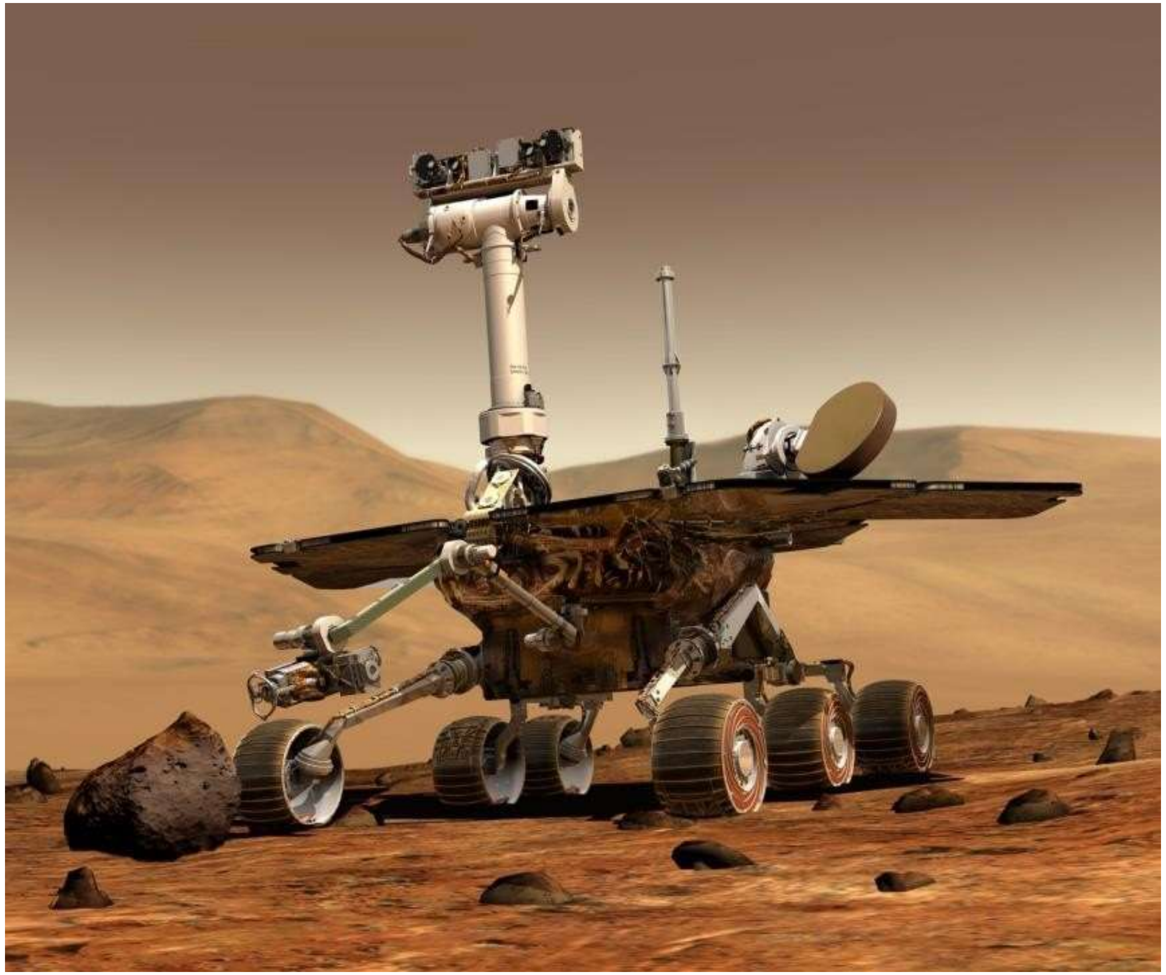
- On average, a person interacts with 100s of embedded systems on daily basis.

APPLICATIONS OF EMBEDDED SYSTEM





EMBEDDED SYSTEMS - EXAMPLES



- NASA's Twin Mars Rovers.
- Microprocessor: Radiation Hardened 20MHz PowerPC From **IBM**
- Commercial Real-time OS.
- Software and OS was developed during multi-year flight to Mars and downloaded using a radio link.

EMBEDDED SYSTEMS - EXAMPLES

Embedded Systems - Examples

Sphero BB-8 From inside

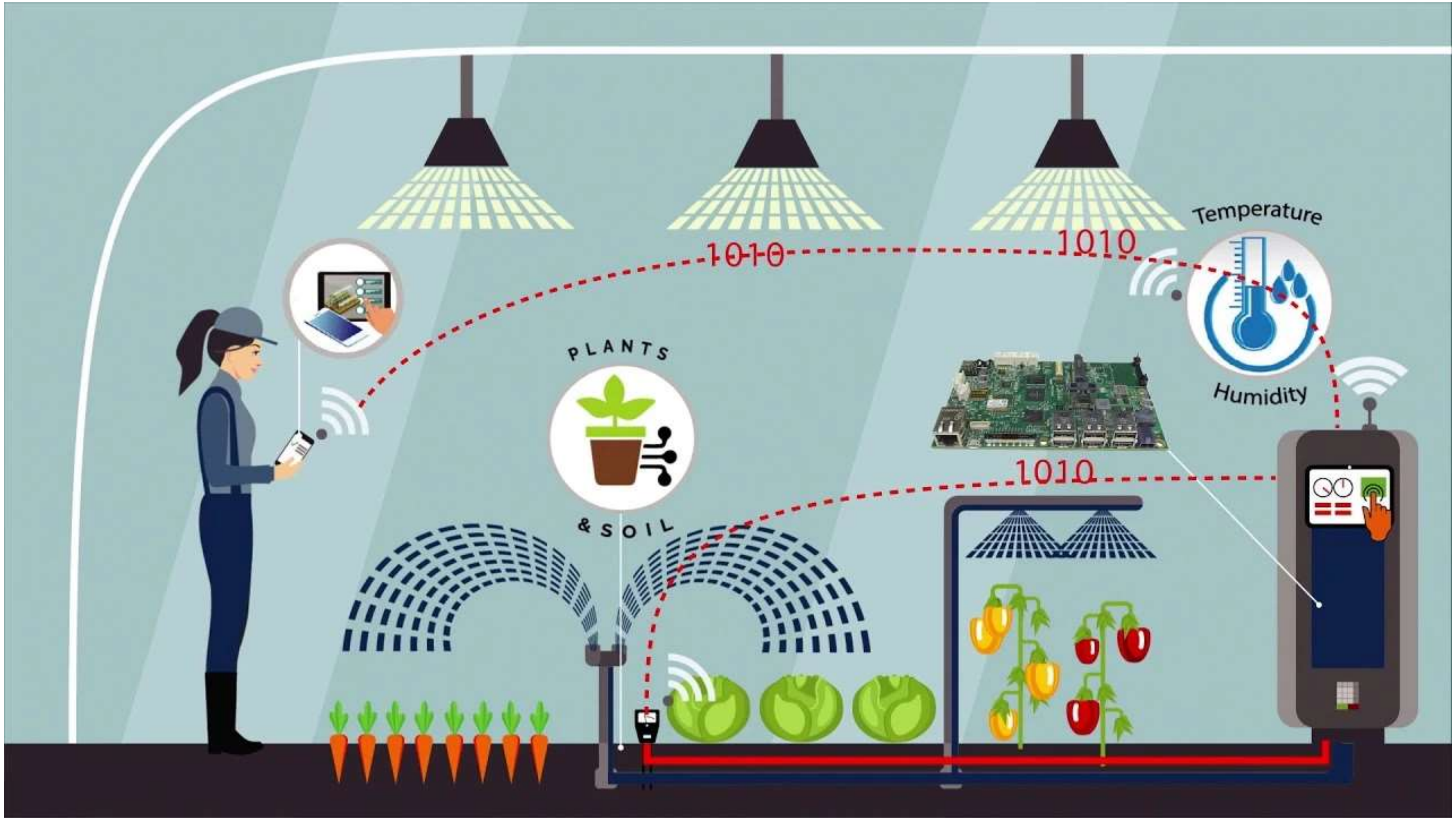
EMBEDDED SYSTEMS - EXAMPLES

Sony Aibo robotic dog
Microprocessor: 64-bits MIPS RISC

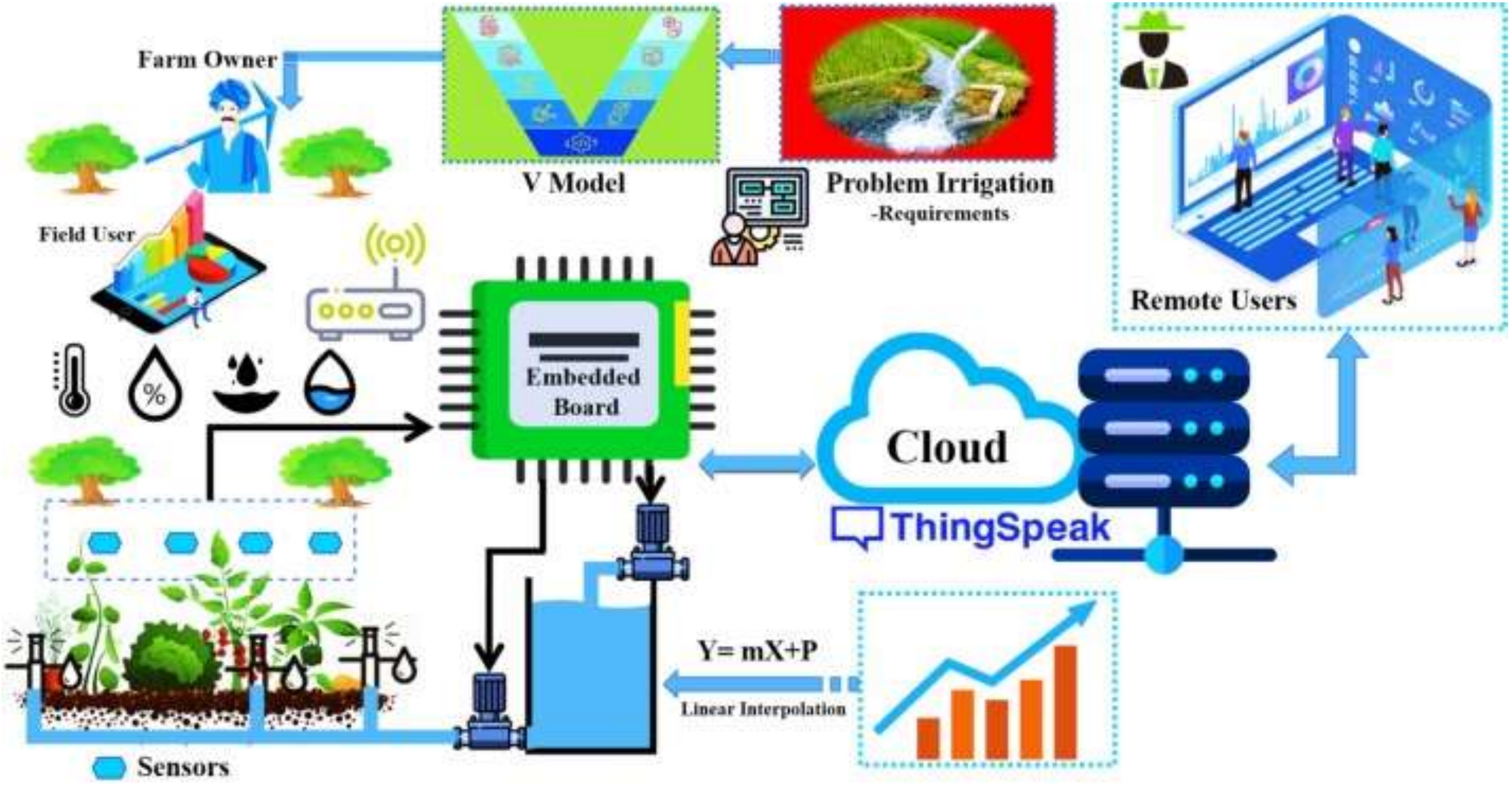
APPLICATIONS OF EMBEDDED SYSTEM

- AGRICULTURAL SECTOR
- AUTOMOTIVE ELECTRONICS
- CONSUMER ELECTRONICS
- INDUSTRIAL CONTROL
- MEDICAL ELECTRONICS

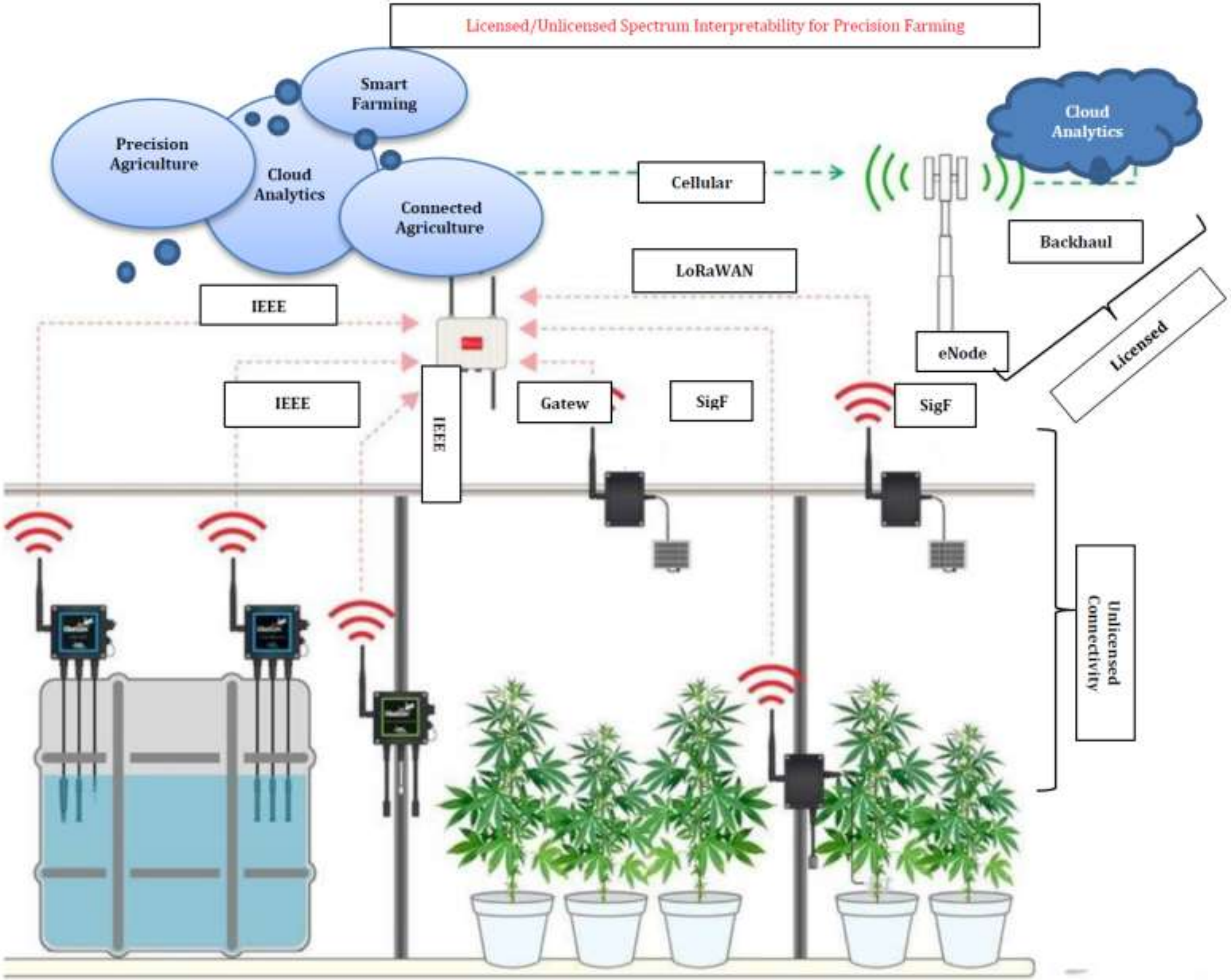
AGRICULTURAL SECTOR – SMART IRRIGATION



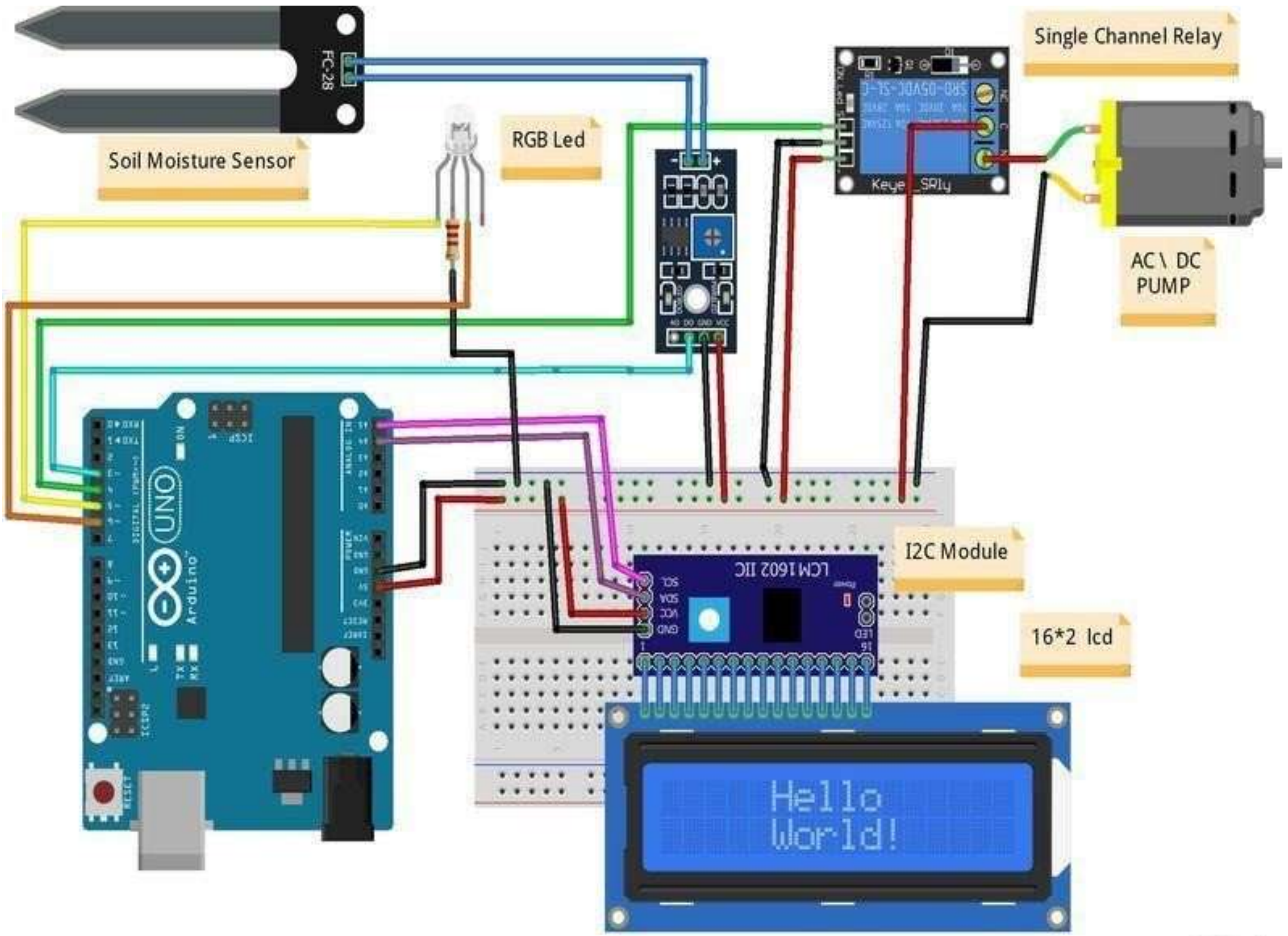
AGRICULTURAL SECTOR – SMART IRRIGATION



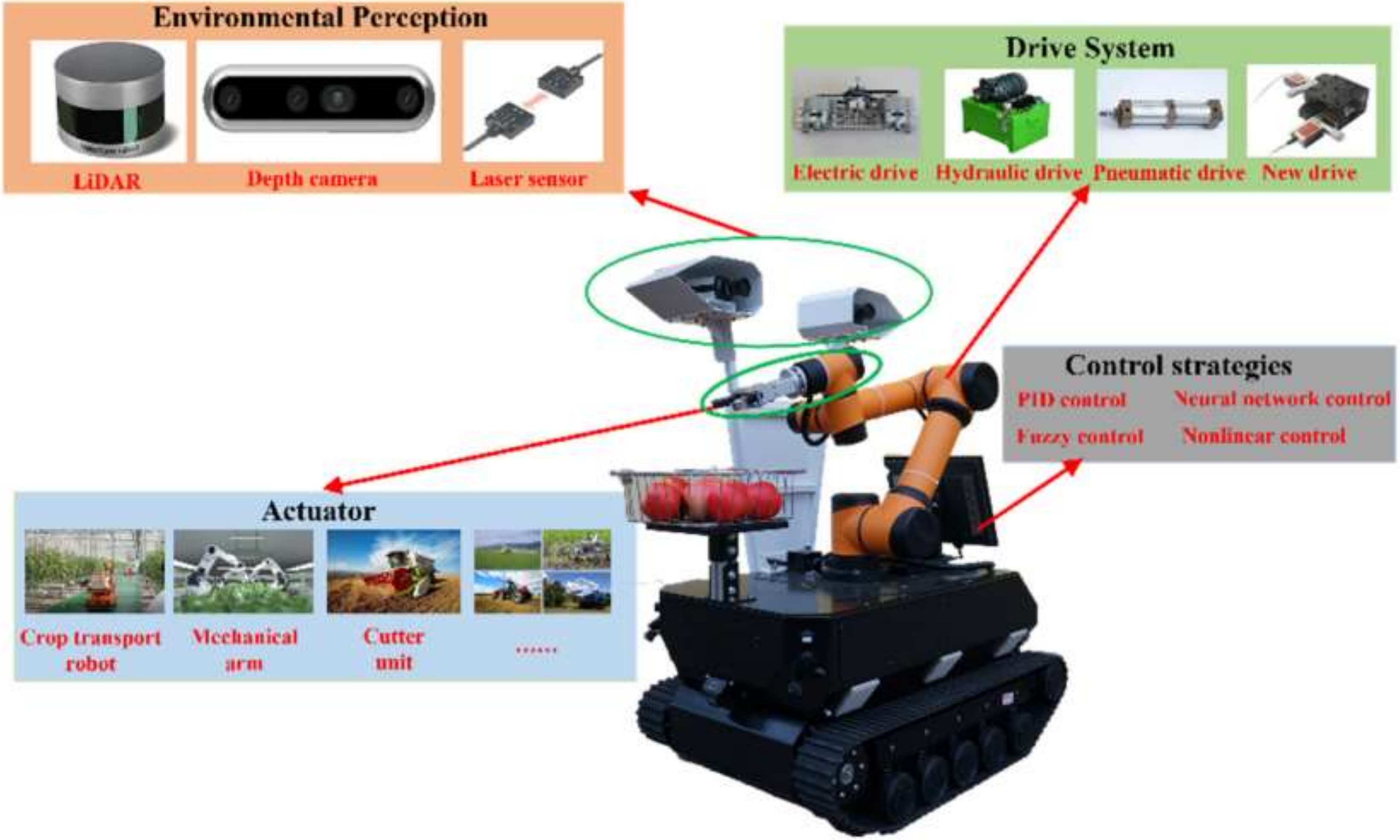
AGRICULTURAL SECTOR – SMART IRRIGATION



AGRICULTURAL SECTOR – SMART IRRIGATION



AGRICULTURAL SECTOR – ROBOT



Xie, D.; Chen, L.; Liu, L.; Chen, L.; Wang, H. Actuators and Sensors for Application in Agricultural Robots: A Review. *Machines* **2022**, *10*, 913. https://doi.org/10.3390/machines10100913

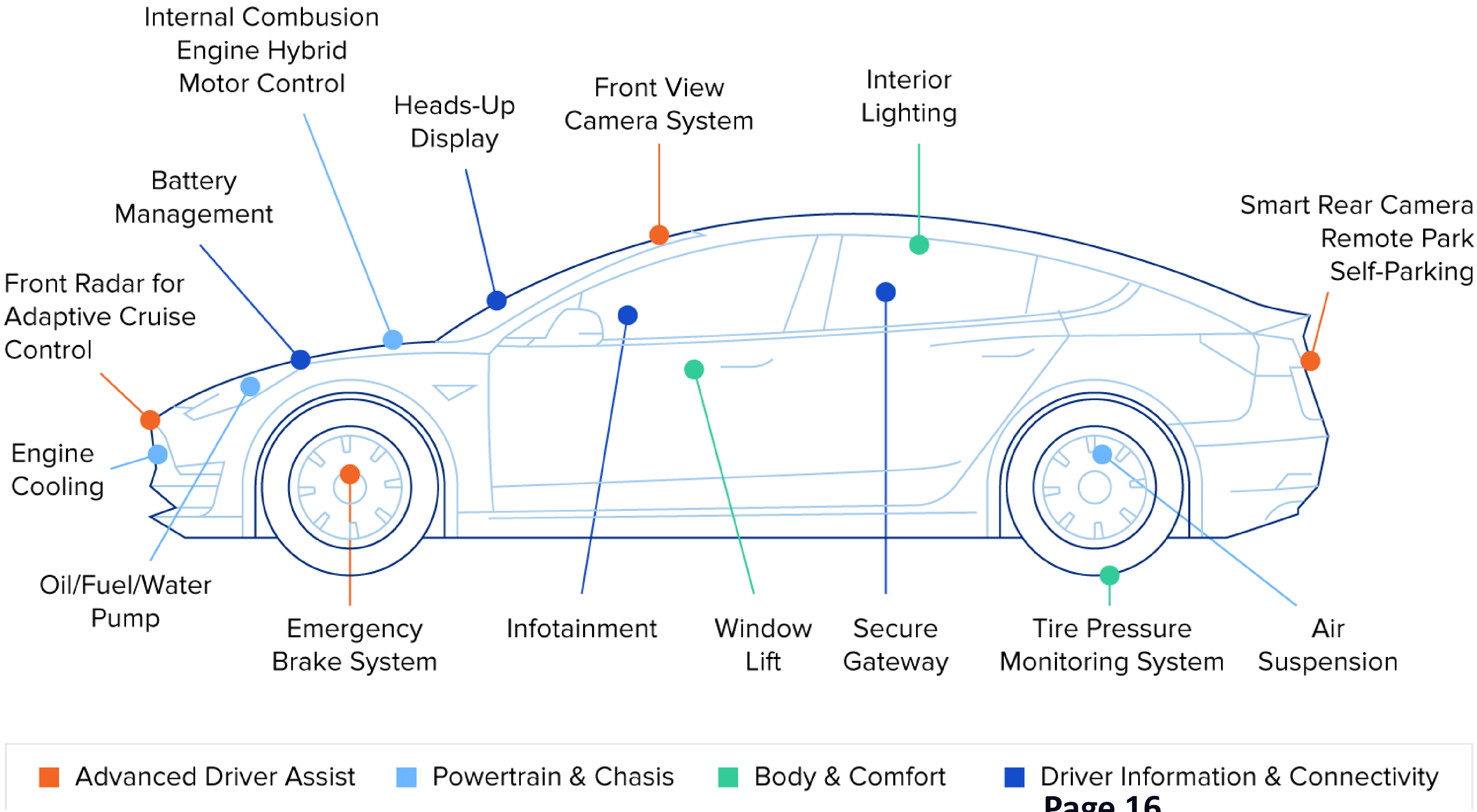
BENEFITS

- Resource optimization
- Increased efficiency and production
- Environmental Sustainability
- Cost saving
- Data-Driven decision making
- Remote monitoring and control Reduce manual labor
- Automated monitoring

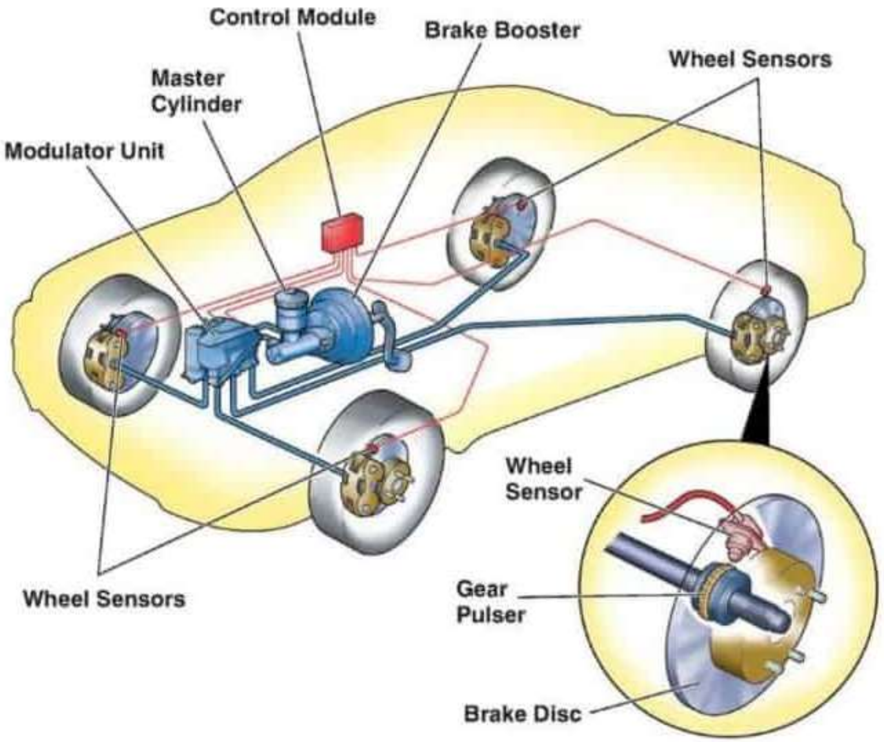
FUTURE TRENDS

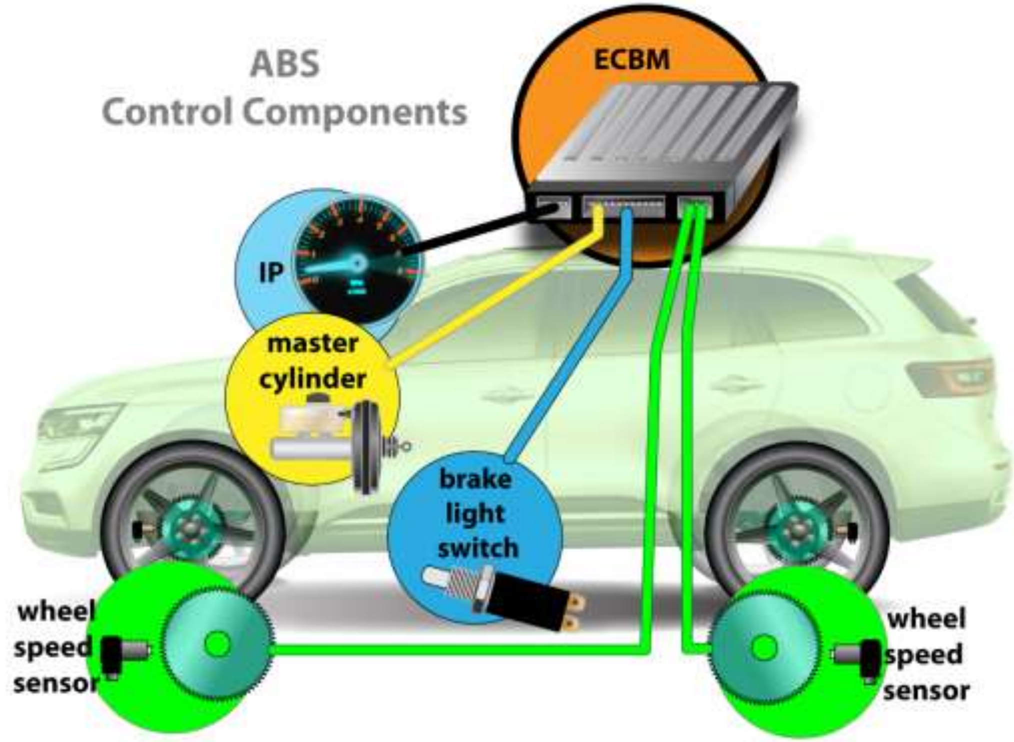
- Artificial intelligence and machine learning
- Robotics and autonomous machinery
- 5G and beyond connectivity into smart farming
- Block chain for supply chain for secure transactions
- Automation in packing and logistics

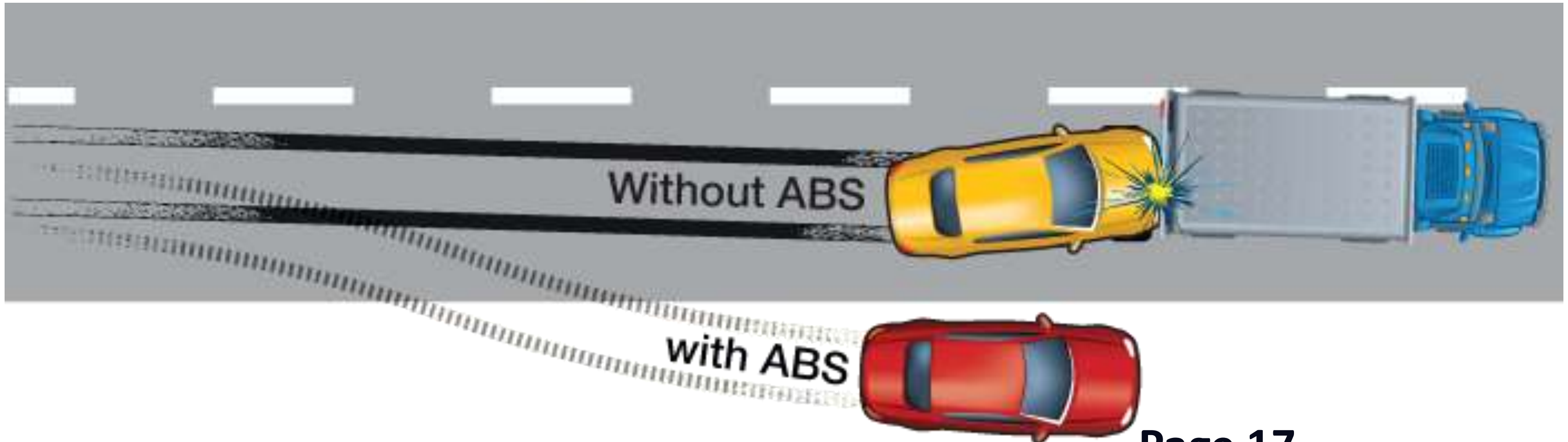
AUTOMOTIVE SECTOR



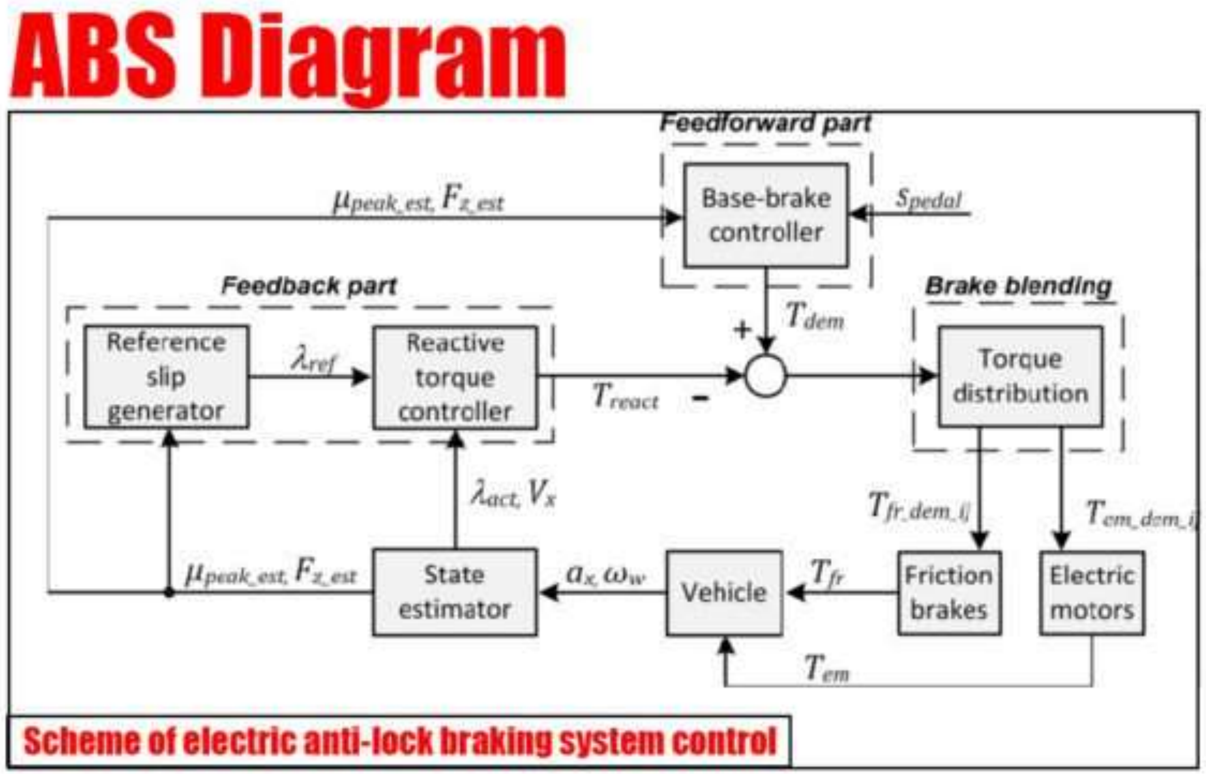
AUTOMOTIVE SECTOR - ABS

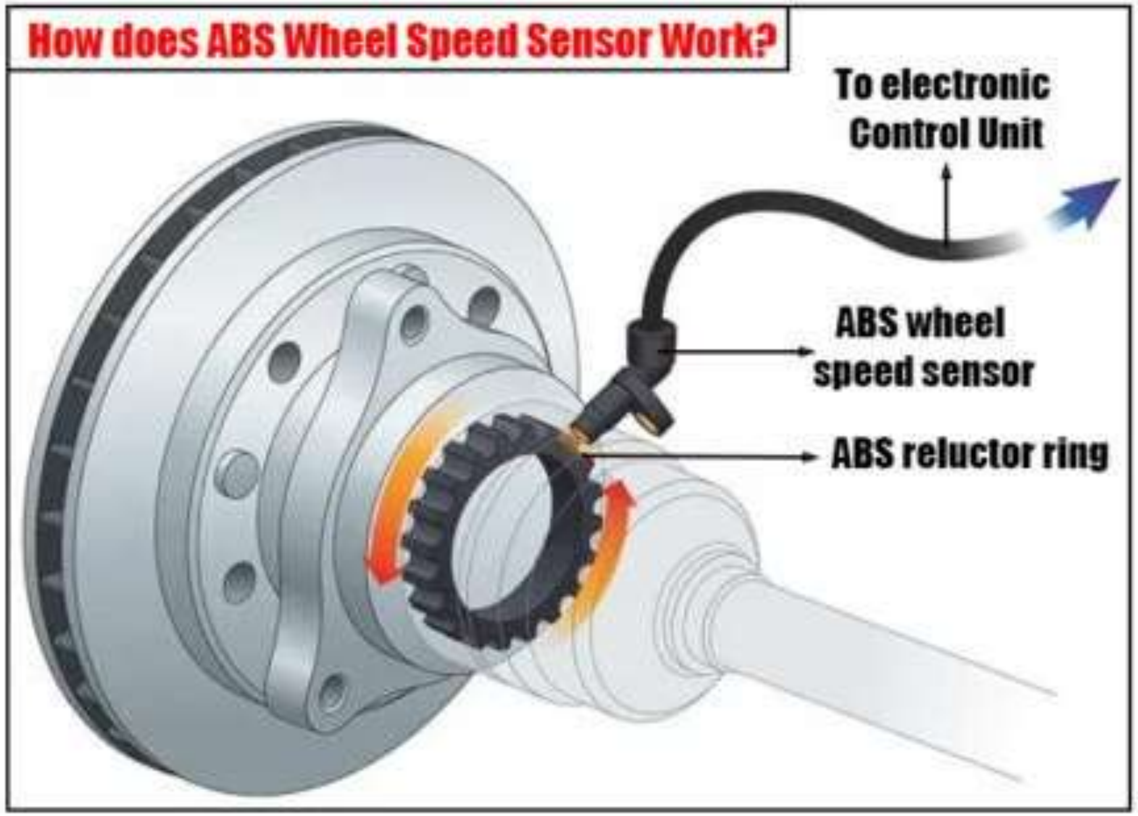


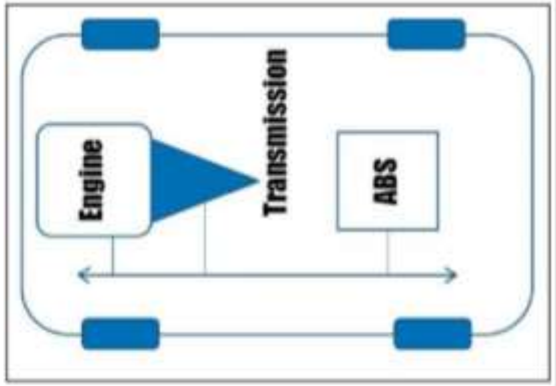


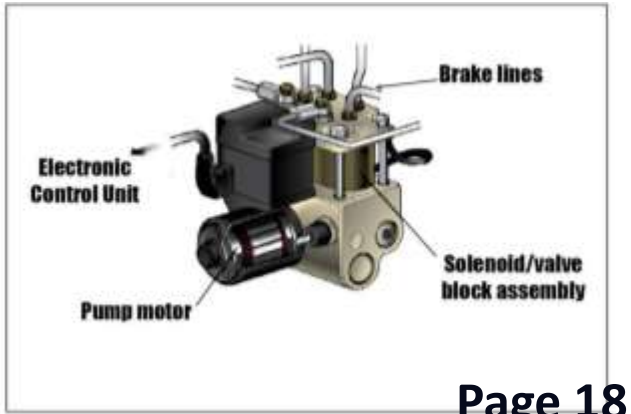


AUTOMOTIVE SECTOR - ABS

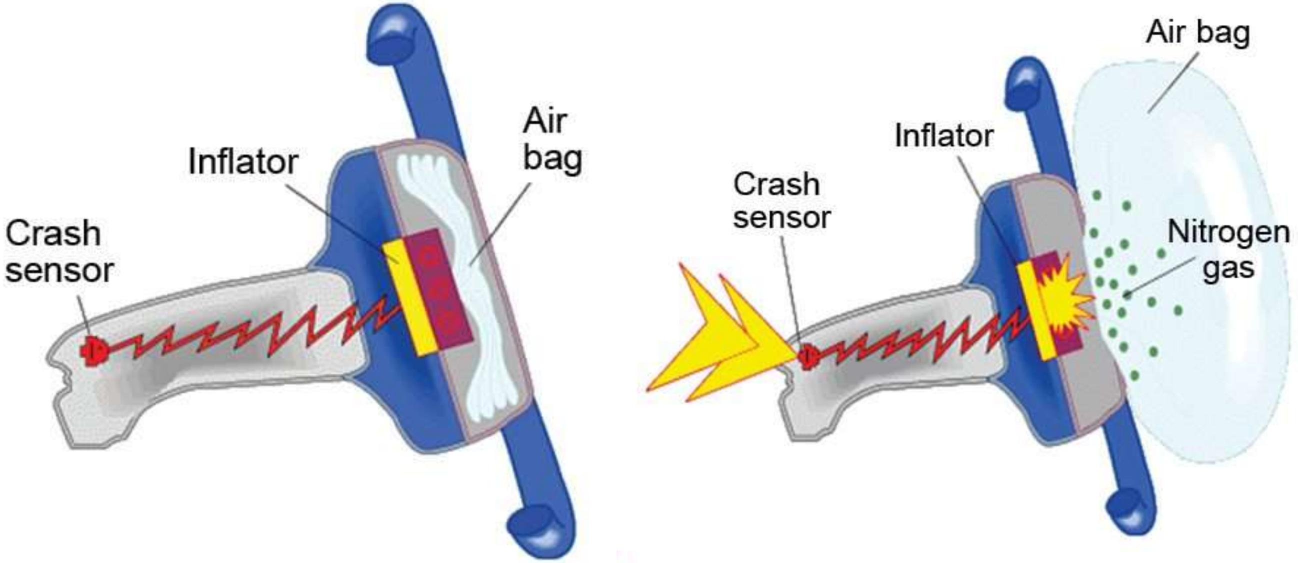


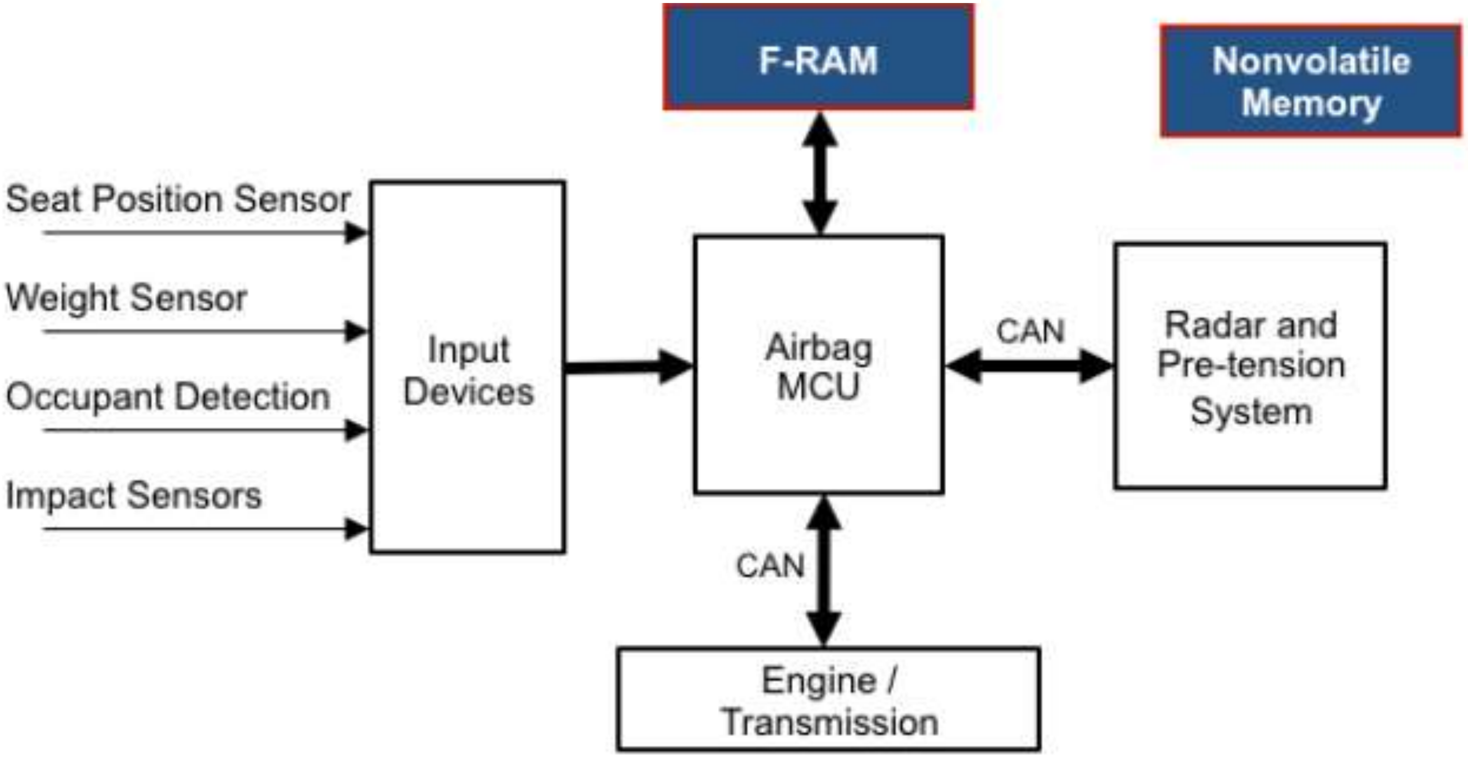




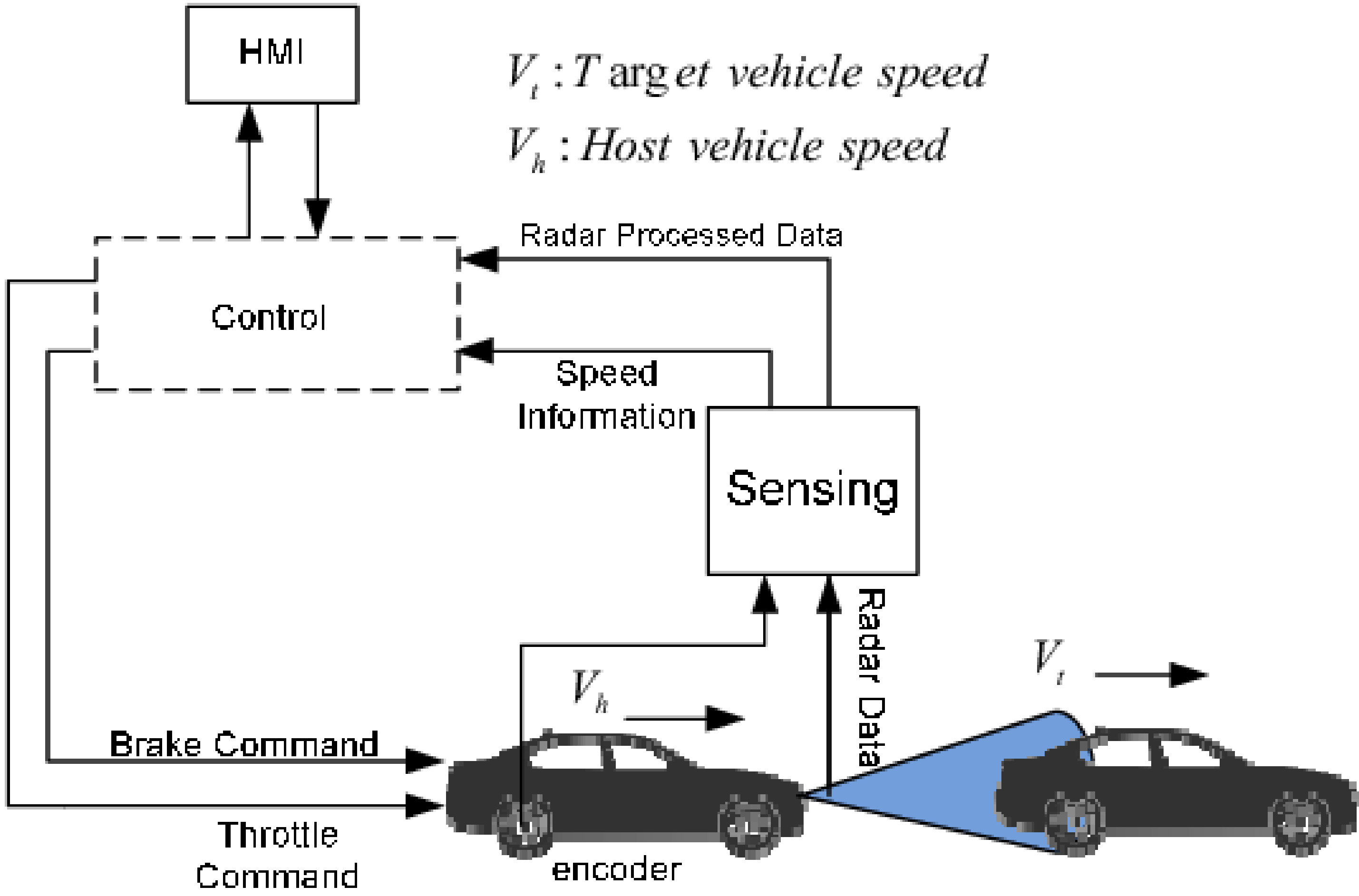


AIR BAG SYSTEM

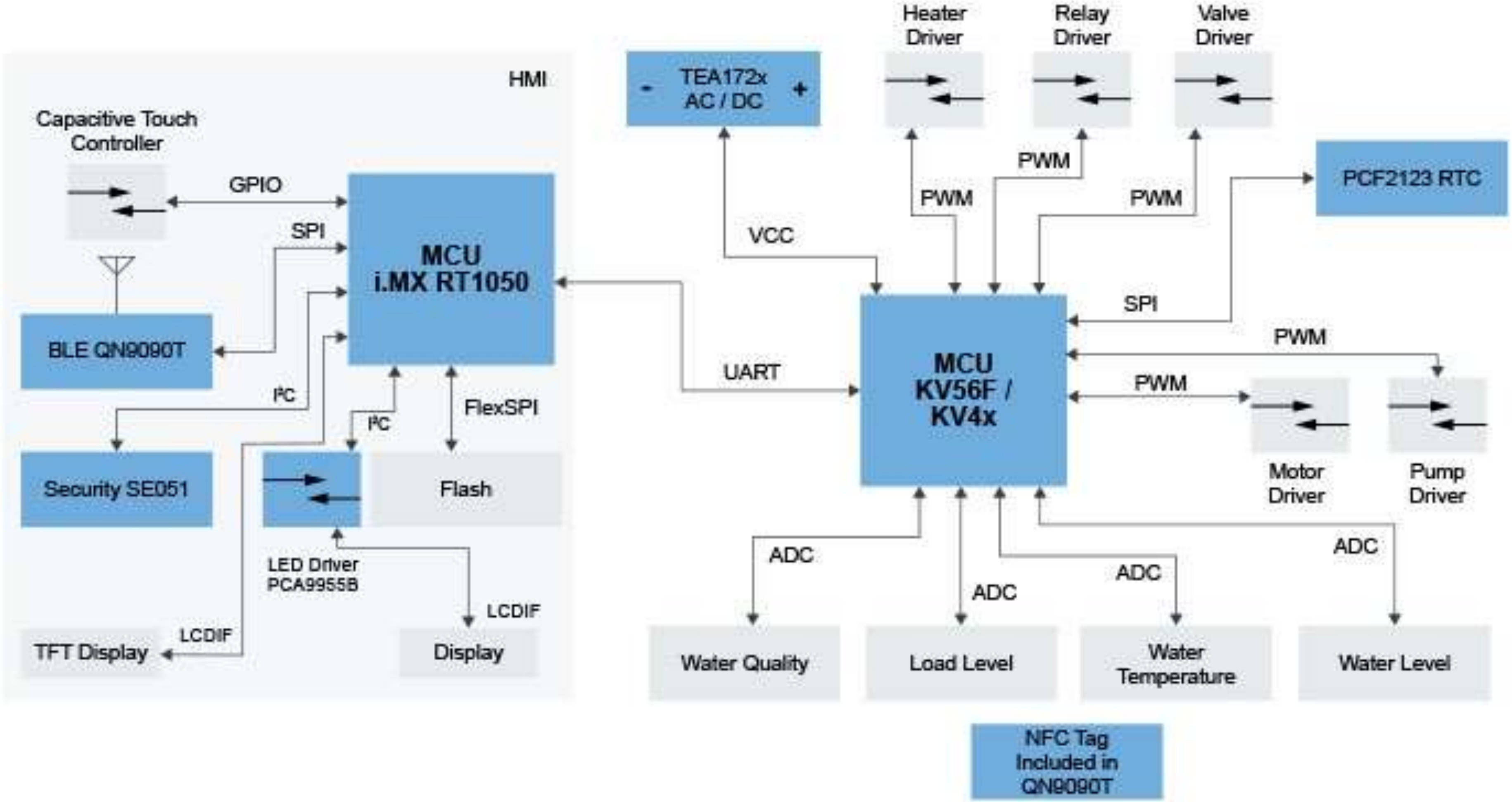


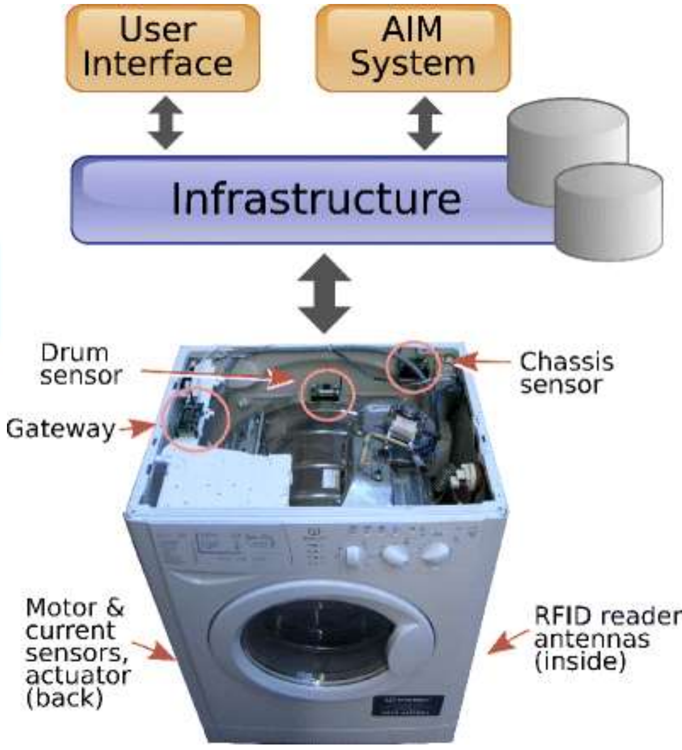


CRUISE CONTROL SYSTEM

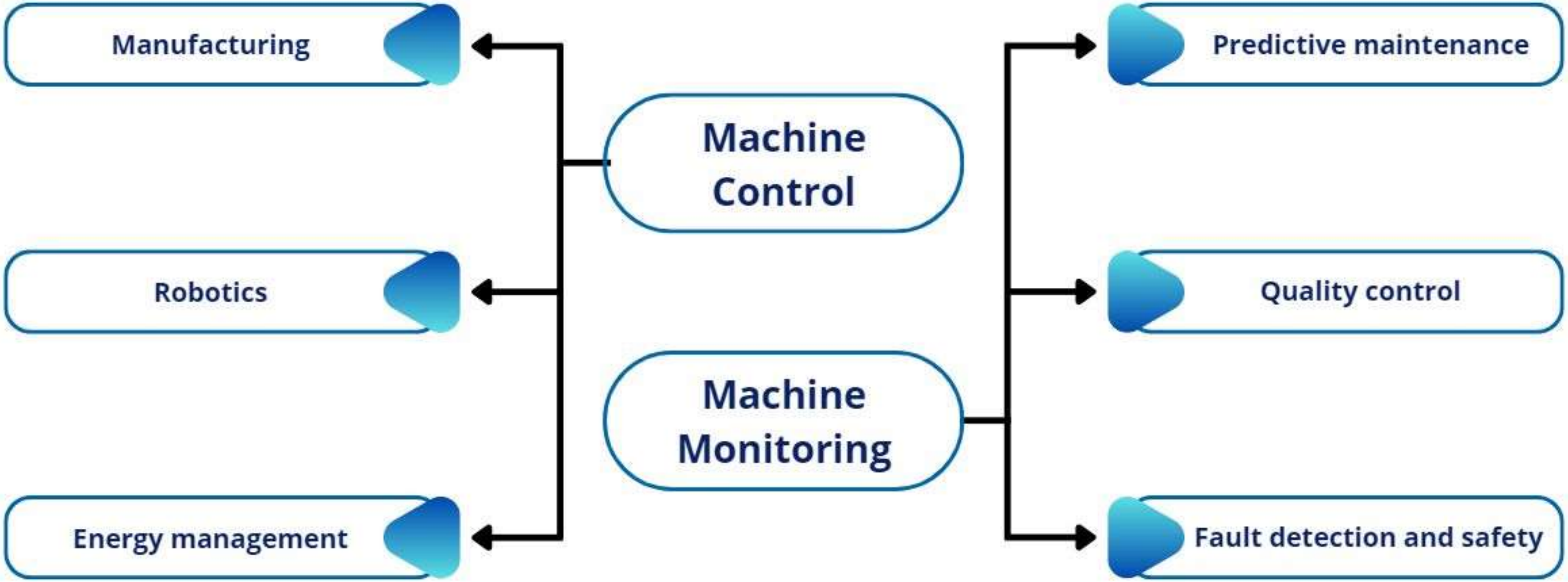


CONSUMER ELECTRONICS- WASHING MACHINE

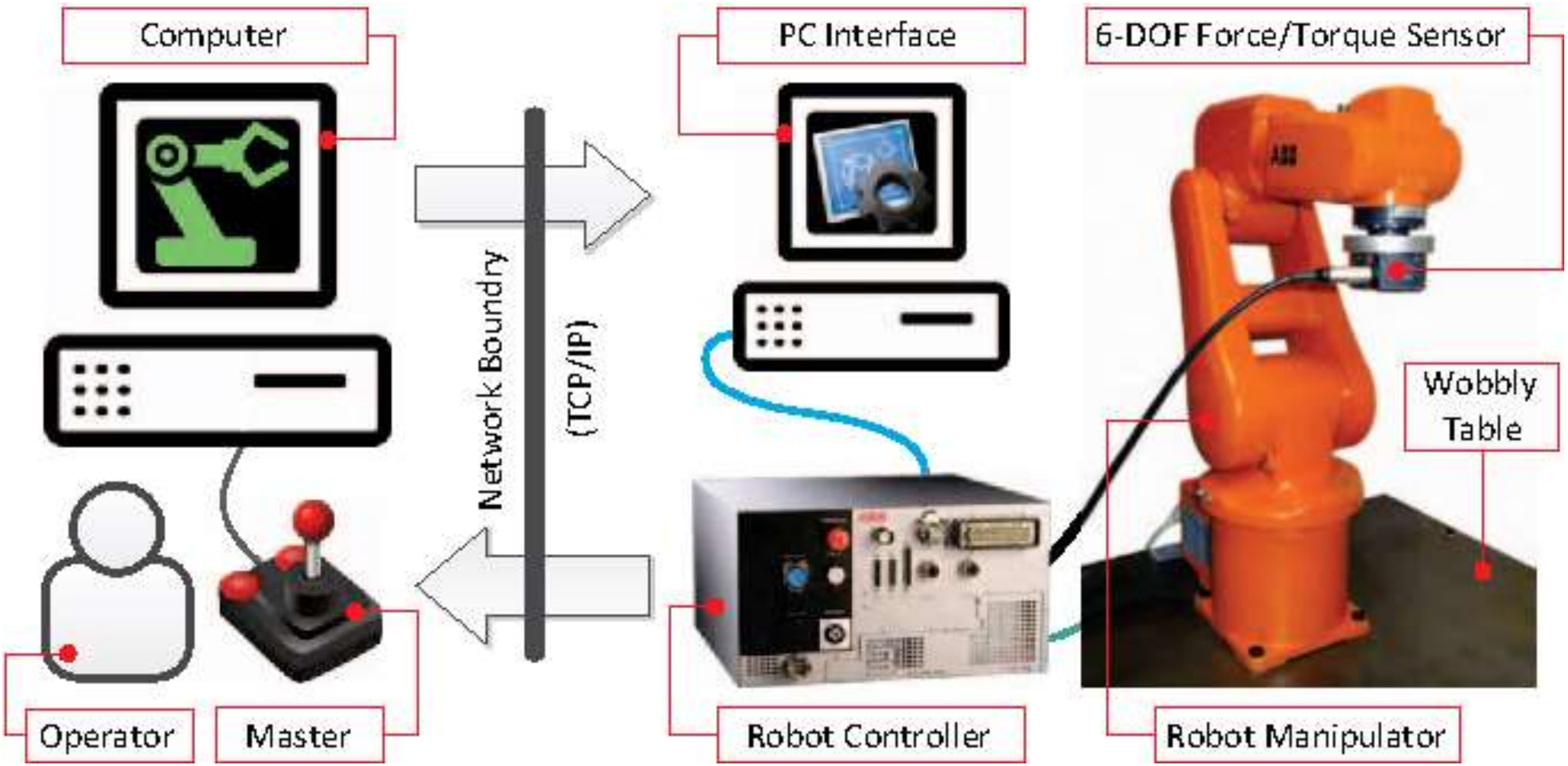




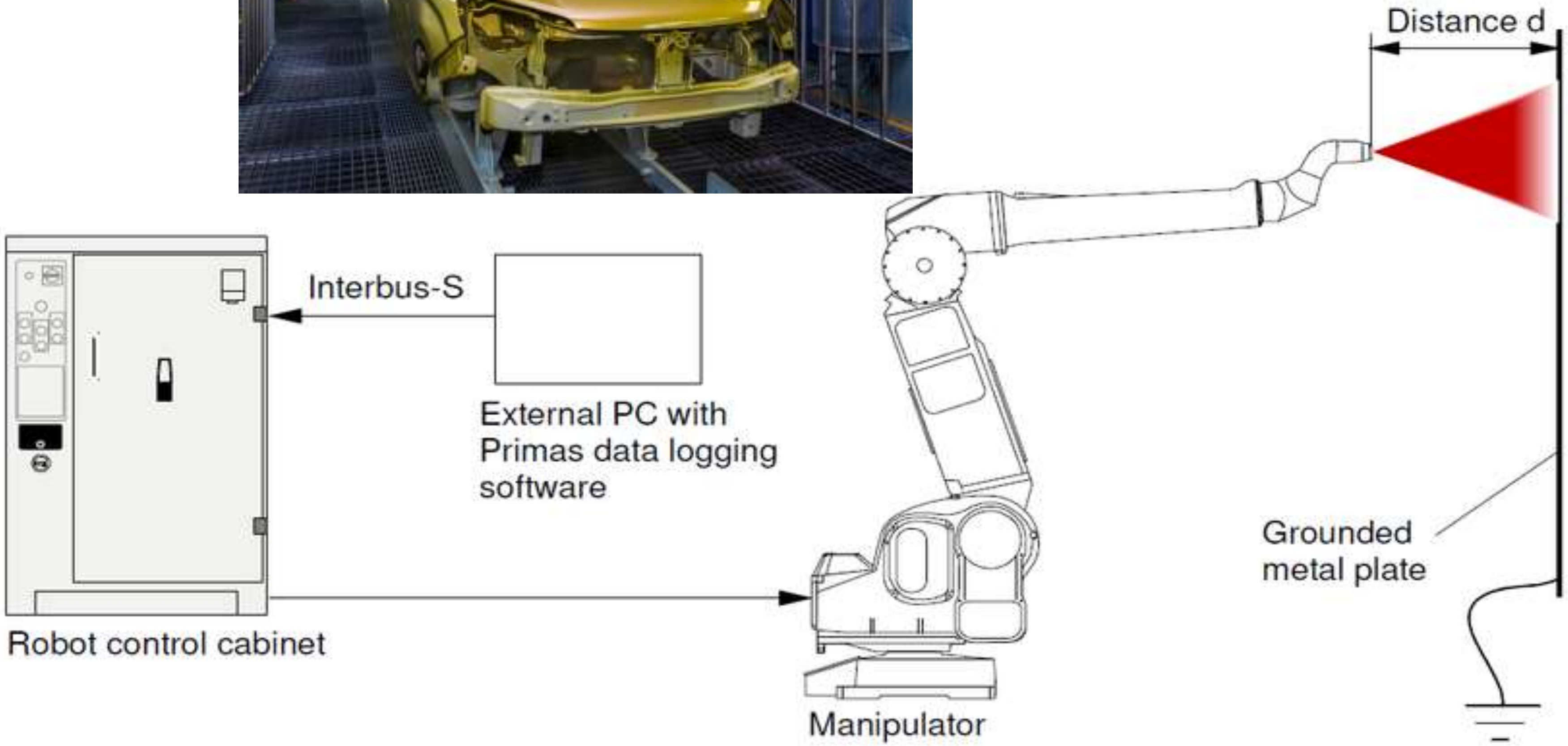
INDUSTRY CONTROL



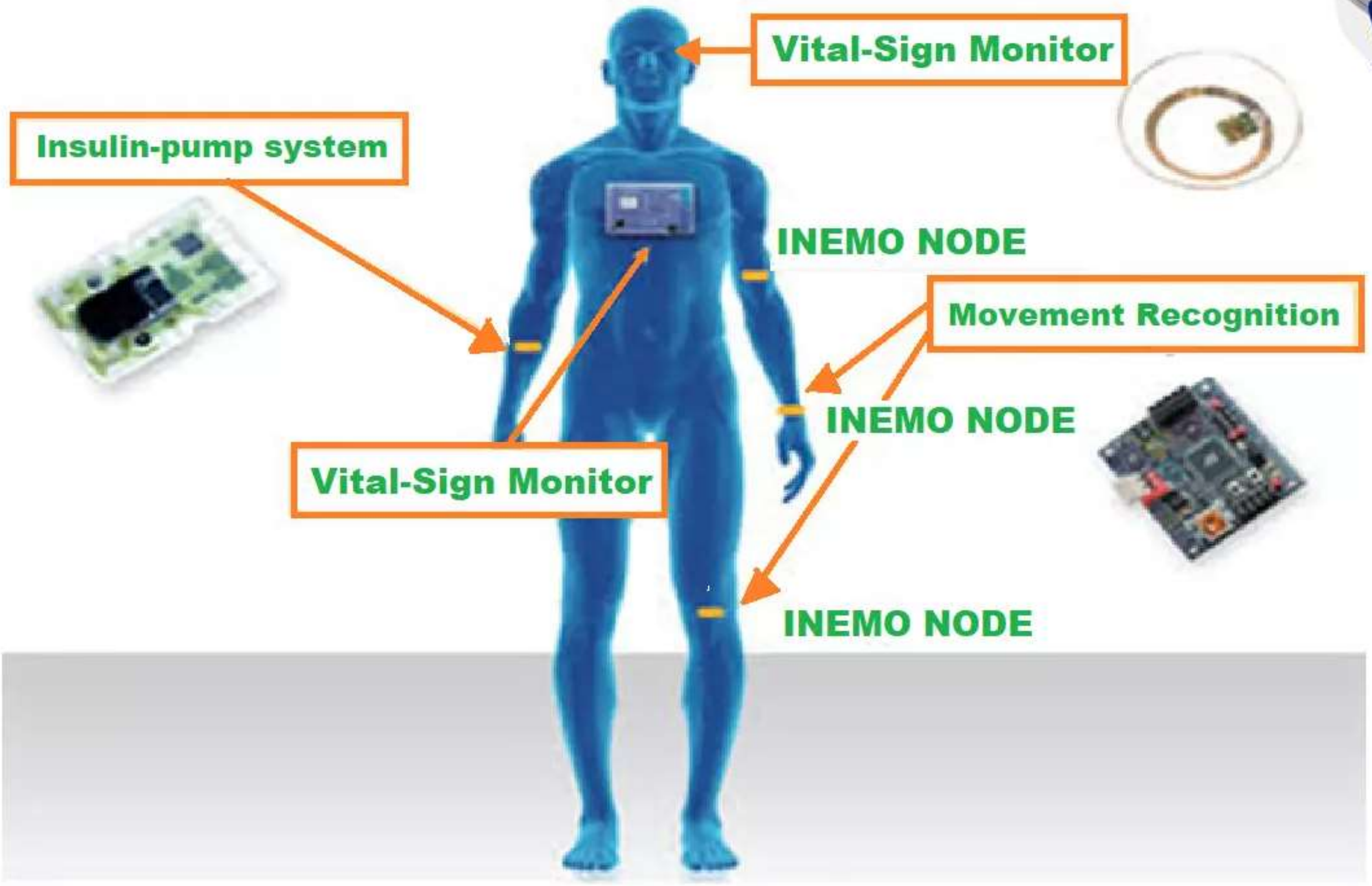
INDUSTRY CONTROL - ROBOTS



INDUSTRY CONTROL - ROBOTS



MEDICAL ELECTRONICS



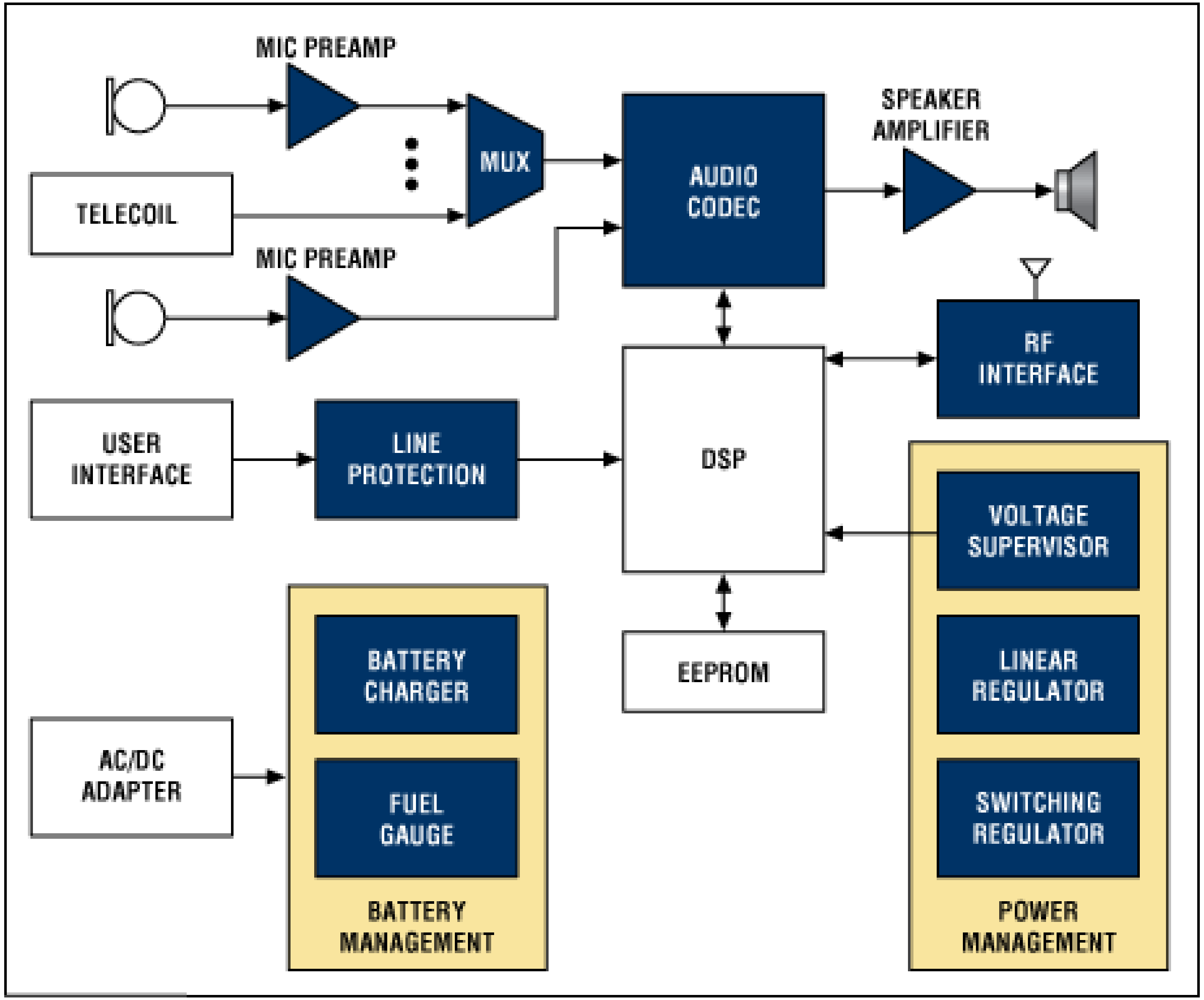
MEDICAL ELECTRONICS

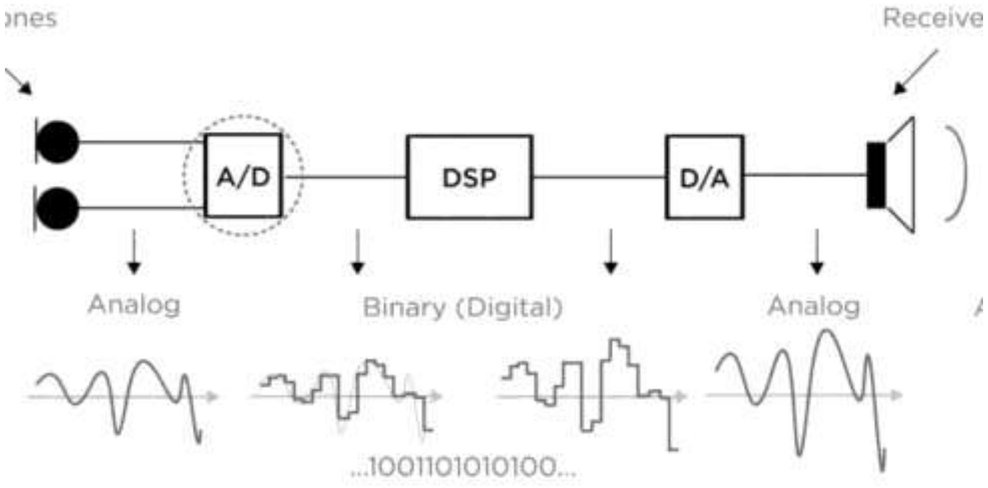
- Pocket calculators
- Hearing aids
- Implantable pacemakers and cardiac defibrillators
- Portable equipment for physically challenged
- Wristwatches
- Wireless computing

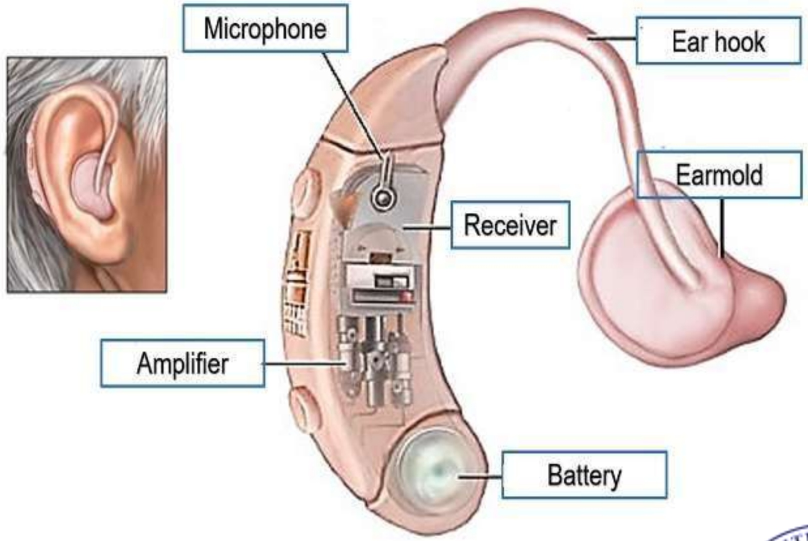
Pacemaker from
Medtronic inc.

Portable defibrillator
From Philips

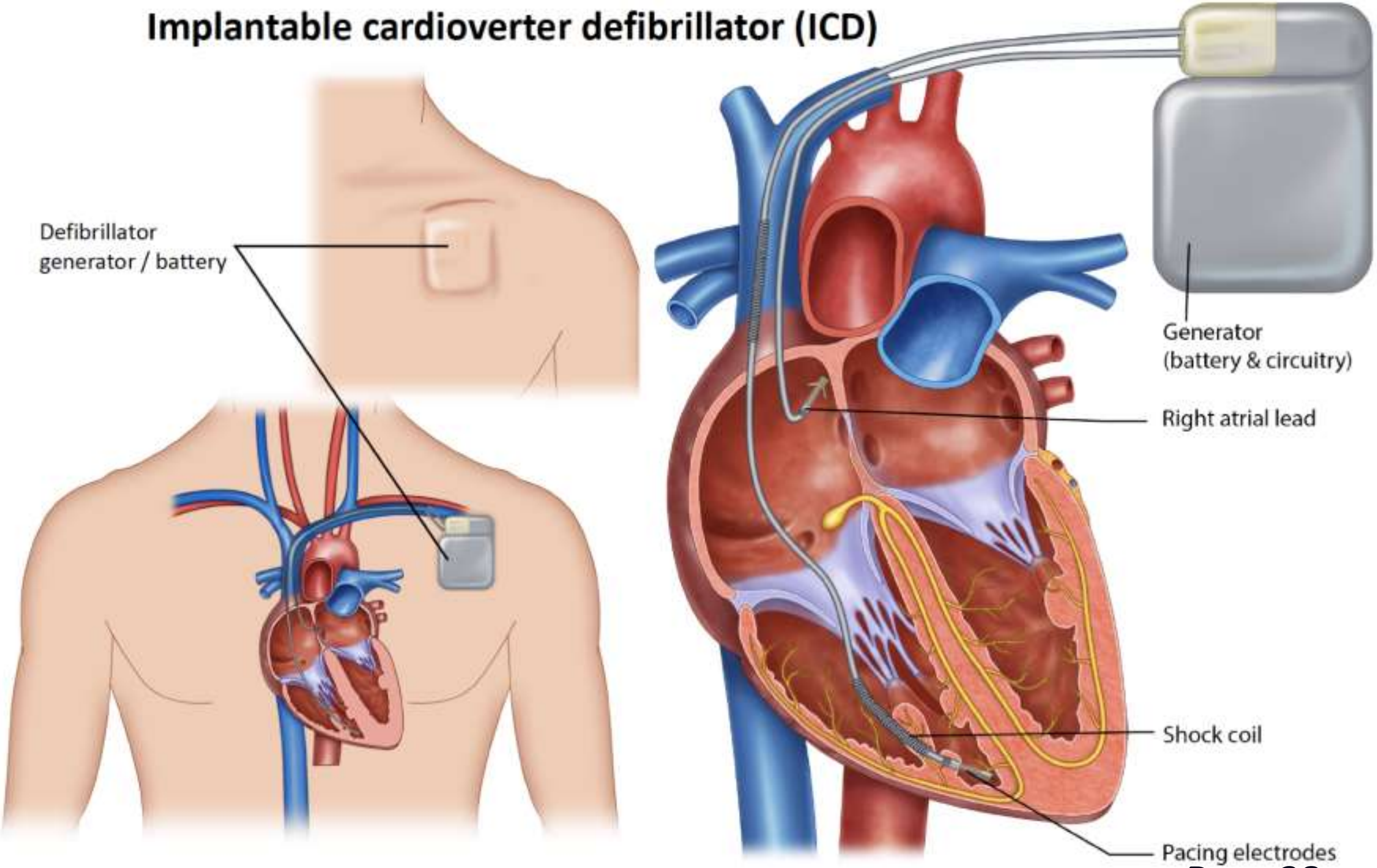
HEARING AID


MIC PREAMP
TELECOIL
MIC PREAMP
MUX
AUDIO CODEC
SPEAKER AMPLIFIER
RF INTERFACE
USER INTERFACE
LINE PROTECTION
DSP
VOLTAGE SUPERVISOR
EEPROM
BATTERY CHARGER
FUEL GAUGE
BATTERY MANAGEMENT
LINEAR REGULATOR
SWITCHING REGULATOR
POWER MANAGEMENT
AC/DC ADAPTER

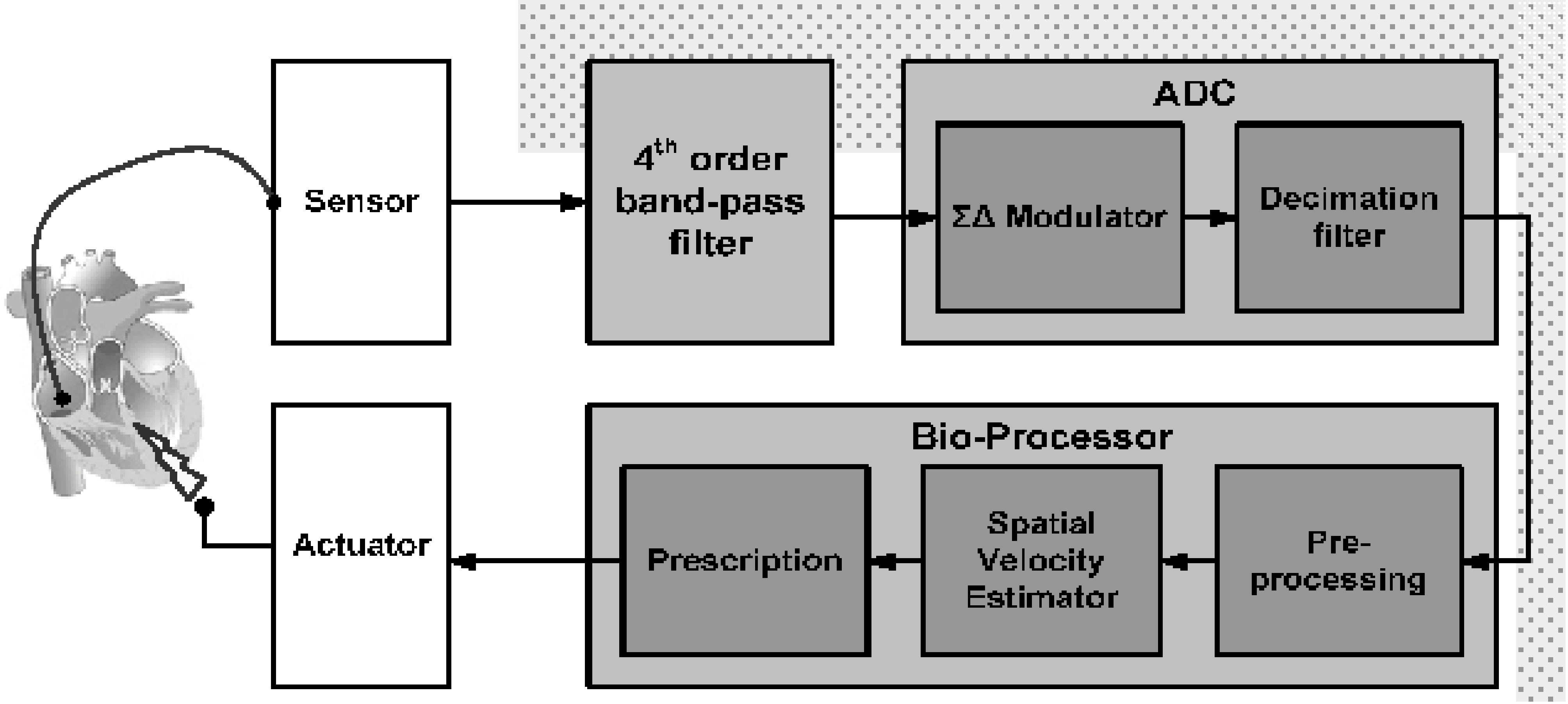

Receive
A/D
DSP
D/A
Analog
Binary (Digital)
Analog
...1001101010100...


Microphone
Ear hook
Earmold
Receiver
Amplifier
Battery

IMPLANTABLE DEFIBRILLATOR



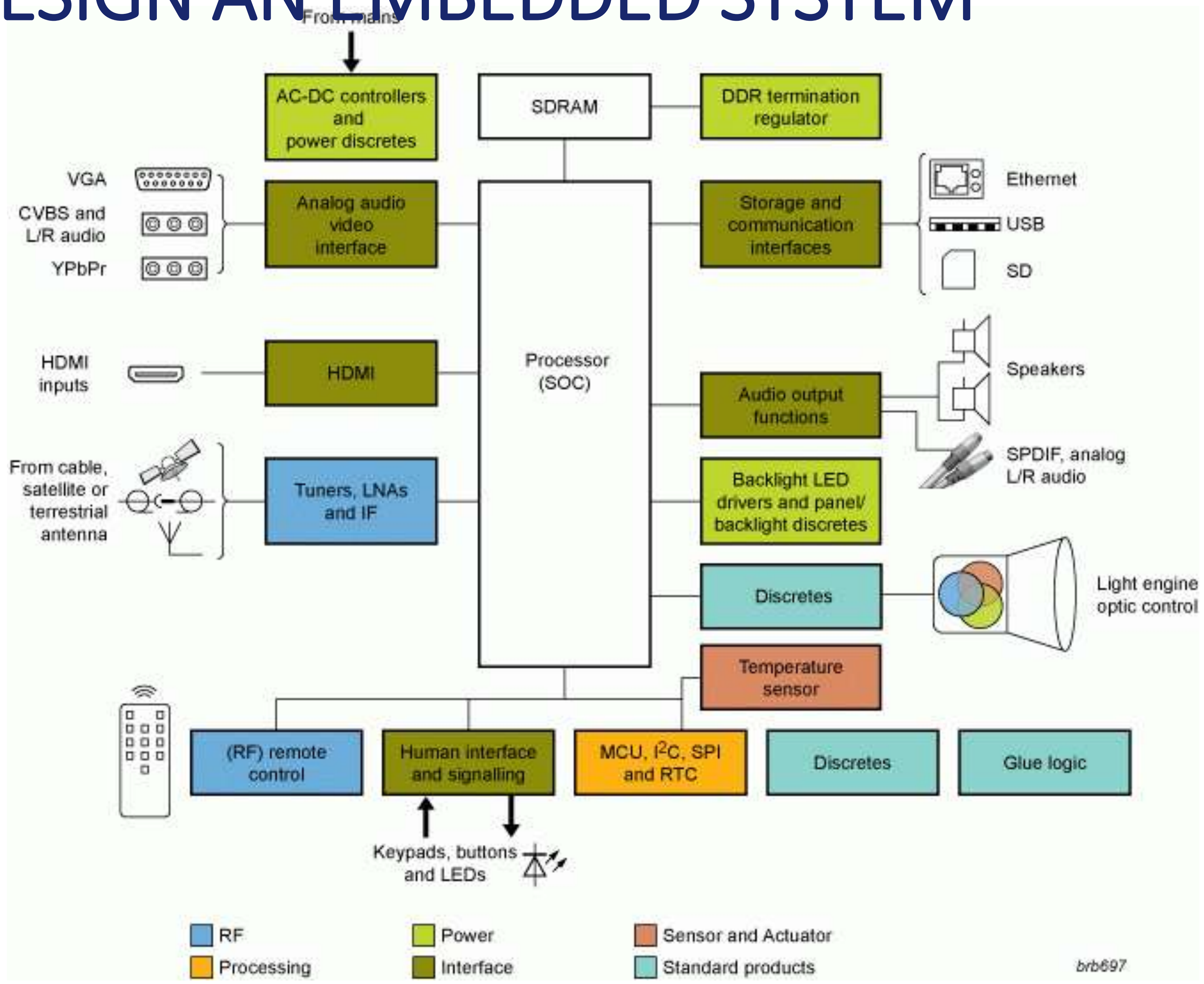
IMPLANTABLE DEFIBRILLATOR



LETS DESIGN AN EMBEDDED SYSTEM

HELP ME TO DESIGN A PROJECTOR ???

LETS DESIGN AN EMBEDDED SYSTEM



HERE IT IS !!

Finishing Note !!

One of the first recognizably modern embedded systems was the Apollo Guidance Computers developed by Charles Stark Draper at the MIT Instrumentation Laboratory used in Apollo Lunar module

It's never too late to build your own !!!